8 Diseases of Coconut

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Introduction

The coconut, *Cocos nucifera*, is the most widely cultivated of all palms. It originated in Melanesia and has been spread by sea currents and man throughout the coastal tropics (Harries, 1992). The genus *Cocos* is monospecific, but many varieties and ecotypes are recognized. Coconut palms are classified as having either a tall or dwarf growth habit. Those of the tall type grow to 30–40 m in height, can live and produce fruit for >80 years, are usually cross-pollinated, and will thrive in diverse soils and environments. Tall palms mature slowly and usually do not bear fruit until they are at least 8 years old. In contrast, dwarfs generally grow no more than 10 m in height and are usually self-pollinated. They begin to bear after 3 years but do not achieve full production until they are >8 years old. Dwarfs are shorter lived than talls and typically have a productive lifespan of 35 years.

Coconut is the archetypal smallholder crop of the coastal tropics (Persley, 1992). It is a low input crop that provides food, drink, fuel, shelter and cash income for producers. Almost all parts of the plant are used. Nut yield varies greatly among varieties, and local selection occurs depending on whether production is for oil or sale on the fresh market as a source of food and drink. Copra, the dried endosperm and the source of edible oil, is the main economic product worldwide. Husks are retted to provide coir for fibre industries, and shells are burnt in kilns to produce charcoal. In some countries, shells and husks are chipped to make a peat mulch substitute that is exported to temperate countries for horticultural use.

Coconuts are rarely grown in pure stands. On smallholder farms, they are under- and interplanted with other crops, including cassava, citrus, coffee, cocoa, maize, mango, millet, sweet potato and spices. In plantation systems, coconuts are often underplanted with coffee and cocoa for which they provide necessary shade (Purseglove, 1972). Breeders are constantly striving to produce higher yielding cultivars that are suitable for a wide variety of tropical environments. Hybrids between talls and dwarfs are favoured by many plantation growers because they tend to mature quickly and are dwarf in stature, which makes nut harvest simpler (Been, 1995).

Coconut is grown on an estimated 11.6 Mha hectares in 86 countries, and ~96% of total world production comes from smallholdings. About 85% of the crop is produced in Asia and the Pacific region. In 1998, world production was ~47.7 million tonnes (Mt), and Indonesia and the Philippines were the leading producers with, 14.7 and 10.5 Mt, respectively (Table 8.1).

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Table 8.1. Coconut production in 1998 in the most important countries in various regions and major diseases that constrain production.[a]

Country	Production (Mt)[b]	Major diseases
Indonesia	14.7	Bud rot, wilt diseases
Philippines	10.5	Cadang-cadang
India	10.0	Kerala (root) wilt
Sri Lanka	2.0	Leaf scorch decline
Mexico	1.1	Lethal yellowing
Brazil	0.65	
Mozambique	0.45	
Tanzania	0.34	Lethal disease

[a]Adapted from Persley (1992). Production data are from the FAOSTAT database http://apps.fao.org/default.htm
[b]Production is in millions of metric tonnes.

Coconut suffers from numerous diseases. The most widespread is bud rot, which is caused primarily by the chromistan, *Phytophthora palmivora*. Bud rot is distributed worldwide, and severe infestations can cause total loss of a crop. Fungi cause most of these diseases, and at least 38 genera have been reported on this host. More than one species may act in concert to reduce the vigour of the palm. For example, in India, a complex of 14 species of fungi, including *Colletotrichum gloeosporioides* and *Exserohilum rostratum*, have been isolated from palms with leaf rot, a syndrome that affects the spear leaf.

Coconut diseases caused by non-fungal agents are often restricted in their geographic distribution. For example, red ring disease, caused by the nematode *Bursaphelenchus cocophilus*, is limited to Central and South America; foliar decay, caused by the coconut foliar decay virus, occurs only on Vanuatu in the New Hebrides archipelago; and cadang-cadang and tinangaja, both caused by viroids, are found only in the Pacific region (Griffiths, 1987; Hodgson and Randles, 1999; Randles *et al*., 1999). Lethal yellowing, caused by a phytoplasma, was first reported from the Caribbean, but diseases attributed to similar phytoplasmas are now recognized in Africa, Central America and Indonesia (Warokka, 1999; Cordova *et al*., 2000).

In this chapter, we describe the most important diseases of coconut. Minor diseases that have limited distributions or small impact on the health of this host are listed in Table 8.2.

The Major Diseases of Coconut

Bud rot and nutfall

Bud rot is a serious disease of coconut that occurs throughout the warm moist tropics.

Symptoms

Bud rot causes a chlorosis and rapid necrosis of the youngest opened leaves (Plate 59). Bending and breaking of affected leaves can occur. Necrotic spotting occurs on leaf bases and pinnae when the youngest leaves are infected in the spindle (Menon and Pandalai, 1960) and hyphal mats of the causal agents may develop on lesions on leaf bases. As infection advances to encompass the newest, unopened spear leaf, it can be pulled out with minimal force, revealing the underlying rotted apical bud. A foul-smelling odour invariably accompanies exposure of these softened and pink to reddish brown tissues (Joseph and Radha, 1975). Characteristically, leaf necrosis progresses quickly throughout the central leaves of the upper crown, leaving a surrounding fringe of unaffected green leaves. In later stages, infection spreads downward into mature, woodier tissues and the upper trunk. Affected areas are water soaked, pinkish and delineated from healthy tissues by a dark border (Fig. 8.1). Although nut fall is not typical of bud rot, inflorescences may become infected, at which time attached nuts abort. Fallen nuts may exhibit superficial water-soaked lesions on their husks from which the causal agents can be readily cultured (Fig. 8.2). Affected palms rarely recover, but when they do they usually have a bitten leaf appearance.

Causal agents

Although *Fusarium moniliforme*, *F. solani* and *Graphium* sp. have been reported as causal agents, species of *Phytophthora* are more com-

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Table 8.2. Miscellaneous diseases of coconut, *Cocos nucifera.*

Disease	Causal agent(s)
Algal leaf spot	*Cephaleuros virescens*
Anthracnose	*Glomerella cingulata* (anamorph: *Colletotrichum gloeosporioides*)
Bacterial bud rot	*Erwinia* spp.
Bitten leaf	*Ceratocystis paradoxa* (anamorph: *Chalara paradoxa*)
Bipolaris leafspot	*Bipolaris incurvata*
Black scorch	*Ceratocystis paradoxa* (anamorph: *Chalara paradoxa*)
Blast	Phytoplasma suspected
Bristle top	Not known
Catacauma leaf spot	*Catacauma mucosum*
Damping-off	*Fusarium* spp.
	Phytophthora spp.
	Pythium spp.
	Rhizoctonia solani
Dry basal rot	*Ceratocystis paradoxa* (anamorph: *Chalara paradoxa*)
Dry bud rot	Not known (possibly vectored by *Sogatella kolophon* and *S. yubana*)
Finschafen disease	Not known
Frond rot	Physiological disorder
Graphiola leaf spot	*Graphiola phoenicis*
Koleroga	*Phytophthora arecae*
Leaf blight	*Cytospora palmarum*
Leaf spots	*Alternaria* sp.
	Botryosphaeria disrupta
	Capitorostrum cocoes
	Cercospora sp.
	Curvularia lunata
	Cylindrocladium pteridis
	Drechslera gigantea
	Epicoccum nigrum
	Exserohilum rostrata
	Helminthosporium sp.
	Macrophoma sp.
	Macrosporium cocos
	Melanconium sp.
	Mycosphaerella palmicola
	Periconiella cocoes
	Pseudoepicoccum cocos
	Phomopsis sp.
	Phyllosticta palmetto
	Ramularia necator
Lixa grande	*Sphaerodothis acrocomiola*
Lixa pequena	*Phyllachora torrendiella*
Powdery mildew	*Oidium* sp.
Pudricion del cogollo	Phytoplasma
Queima das folhas	*Botryosphaeria cocogena* (anamorph: *Diplodia theobromae)*
Root rot	*Fusarium* spp.
	Phytophthora spp.
	Pythium spp.
	Rhizoctonia solani
Soccoro wilt	Not known
Stem necrosis	Phytoplasma suspected
Stigmina leaf spot	*Stigmina palmivora*
Thread blight	*Pellicularia filamentosa*, *P. koleroga*
	Corticium penicillatum

From Ploetz *et al.* (1999).

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Fig. 8.1. Cross-section of coconut apex affected by bud rot. The affected areas are water-soaked, pinkish and delineated from healthy tissues by a dark border (photo: J.J. Ooka and J.Y. Uchida).



Fig. 8.2. Coconut from a tree affected by bud rot. Note the water-soaked, darkened areas at the stem end of the nut (photo: J. Flood).

mon. *P. palmivora* is most significant, and was first isolated from affected palms in India by Butler (1906). It and *P. nicotianae*, a less frequent cause of the disease, are described in Chapter 1. *P. heveae*, cause of bud rot and nutfall in Côte d'Ivoire, is described in Chapter 3 under Phytophthora cankers.

An unnamed species of *Phytophthora* (formerly referred to as *P. katsurae*) was described in Hawaii (Uchida *et al*., 1992). It causes bud and fruit rot on Hawaii, Kauai, Maui and Oahu. It produces sporangia on artificial media that are 31 (23–42) × 40 (31–49) μm, non-caducous, papillate, ovoid and bilaterally symmetric. It is homothallic and produces oogonia with varying numbers of irregular surface protuberances and tapered, funnel-shaped bases. They are 27 (22–31) μm in diameter, antheridia are amphigynous, and oospores are aplerotic. The species has longer sporangia than *P. katsurae* (40 versus 28 μm) and fewer oogonial protuberances.

A fifth species, *P. arecae*, causes primarily nutfall and mahli (nut and flower) disease in India and Sri Lanka (Erwin and Ribiero, 1996). Chlamydospores are uncommon and 35–40 μm in diameter. Sporangia are 40–50 × 35–40 μm, broadly ellipsoid to nearly spherical, papillate and caducous, with 1–6 μm long pedicels (Fig. 8.3) (Stamps, 1985). The pathogen is heterothallic. Oogonia average 30 μm in diameter, and antheridia are amphigynous, 14 × 15 μm, and often wider than long.

Epidemiology

Bud rot affects palms of all ages, but those between 15 and 40 years old are most susceptible. It occurs in all coconut-growing regions, but is most severe in areas with high rainfall (Cook, 1971). Damage caused by high winds and hurricanes frequently is associated with the disease, and whether mature trees can be affected in the absence of such injury is debated. Spread occurs primarily via rainsplashed zoospores and caducous sporangia (*P. palmivora* and *P. heveae*). Oospores and chlamydospores can survive in soil and coconut debris such as cull piles, and infected roots are also reservoirs of inoculum.

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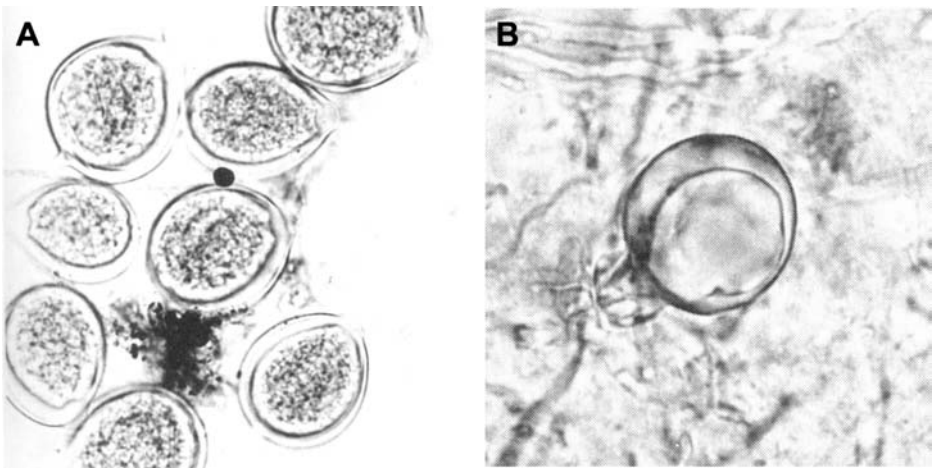


Fig. 8.3. (A) Sporangia and (B) an oospore of *Phytophthora arecae* (from CMI description no. 833).

Management

Few control measures are successful. Good cultural conditions are an essential first step. Infested debris and dead trees should be removed from plantations and nurseries and burned. To avoid prolonged periods of free moisture on host surfaces, nurseries should not be irrigated at night. Thevenin *et al.* (1995) reported that bud rot could be controlled by a stem injection of 20–25 ml of Foli-r-fos 400AS per palm at the onset of the rainy season.

Cadang-cadang and tinangaja

Cadang-cadang is the major cause of death of coconut palms in the Philippines. Although cadang-cadang-like symptoms were described as early as 1914 (Randles *et al.*, 1997), it was not until the 1950s and 1960s that the disease began to reach epidemic proportions. Randles *et al.* (1997) calculated losses due to cadang-cadang of US$39 million for 1978 and US$21 million for 1980.

The disease is confined to, and continues to spread on, Catanduanes, Masbate and Northern Samar Islands and the Bicol Peninsula of Luzon Island (Hanold and Randles, 1997). A similar disease, tinangaja, was reported from Guam in 1917, and by 1946 had destroyed the coconut industry on that island (Boccardo, 1985).

Causal agents

Cadang-cadang and tinangaja are caused by viroids, small, naked, single-stranded circular RNAs. The *Coconut cadang-cadang viroid*, CCCVd, was first isolated by Randles (1975). It has a basic 246 nucleotide structure consisting of a central conserved region of 44 nucleotides that is common to many viroids (Haseloff *et al.*, 1982). It predominates in the early stages of infection, but is replaced by larger forms (287–301 nucleotides) as the disease progresses to the mid- and late stages. This variation is due to the reiteration of 41, 50 or 55 nucleotides. The *Coconut tinangaja viroid*, CtiVd, has a sequence of 254 nucleotides with a 64% homology with the basic CCCVd molecule (Keese *et al.*, 1988). Differences in the sequences of CCCVd and CTiVd could account for the variation in symptoms of the respective diseases.

Symptoms

There are three main stages in the development of cadang-cadang, early (E), mid (M) and late (L) (Fig. 8.4) (Randles, 1997). At the E0 stage, CCCVd is detected in the youngest leaves, but plants are symptomless. After 1–2 years, the E1 stage develops as newly formed nuts become more rounded and exhibit equatorial scarifications, but no leaf symptoms (Fig. 8.5). At E2,

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Fig. 8.4. Coconut palms in various stages of decline caused by the *Coconut cadang-cadang viroid* (CCCVd). Note the understorey of *Alpinia* sp., a symptomless host of CCCVd variants (photo: J.W. Randles).

more nuts are rounded and scarified, chlorotic spots appear on leaves, and inflorescences are stunted with tip necrosis and the loss of some male florets. By E3, leaf spots enlarge, fewer nuts are produced, and new inflorescences are stunted and sterile. Spathe, inflorescence and nut production declines and then ceases by the M stage. Leaf spots become more numerous, but appear only on leaves at or lower than positions 3 or 4 in the crown. By the L stage, leaves decline in size and number, and leaflets become brittle. Leaf spots coalesce, resulting in a general chlorosis, crown size is reduced, and ultimately the palm dies. The length of time that it takes for a palm to die varies with age. In 22-year-old palms, this averages 7.5 years, whereas in 44-year-old palms the average is 15.9 years (Zelazny and Niven, 1980).

Symptoms of tinangaja differ slightly from those of cadang-cadang (Hodgson and Randles, 1999). The earliest recognizable symptom is a reduction in canopy size, but this may be missed until yellow spots appear on the leaves and small elongate nuts that lack a kernel are produced (Fig. 8.6). As the disease progresses, palms decline, the crown shrinks and the upper trunk may show tapering. Ultimately, nut production ceases and the palm dies.

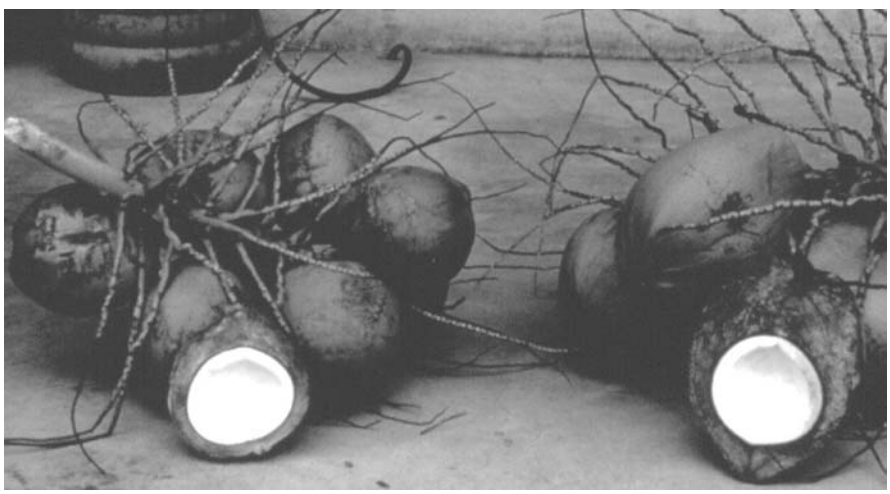


Fig. 8.5. Nuts from healthy (right) and cadang-cadang-affected coconut palms (photo: J.W. Randles).

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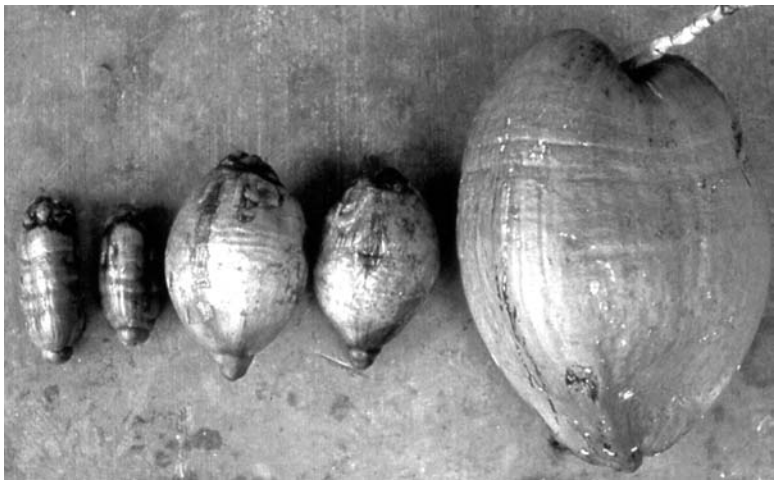


Fig. 8.6. Spindle-shaped nuts from a tinangaja-affected coconut palm compared with one from a healthy palm (right) (photo: J.W. Randles).

Epidemiology

CCCVd has induced symptoms typical of cadang-cadang after artificial inoculation of healthy palms (Randles *et al*., 1977; Mohamed *et al*., 1985). The experimental host range of CCCVd is narrow and limited to the following palm species: *Adonidia merrilli*, *Areca catechu*, *Coryphya elata*, *Dypsis lutescens*, *Elaeis guineensis*, *Phoenix dactylifera*, *Ptychosperma macarthuri* and *Roystonea regia* (Imperial *et al*., 1985). Experimentally infected palms all develop yellow leaf spots and often are stunted.

Production from affected palms falls below economic levels ~5 years before they die (Price, 1971). Studies on the epidemiology of cadang-cadang and tinangaja have been constrained by several problems. These include a long latent period between infection and symptom appearance; difficulties in recognizing early stages of the diseases; no knowledge of how natural transmission of the pathogens occurs; and low rates of disease spread. All of these factors have conspired against the development of sustainable control measures.

Although there have been many studies of the spread of cadang-cadang, few have been conducted on tinangaja since suitable diagnostic tools for it were developed only recently (Hodgson and Randles, 1998, 1999). The rate and pattern by which cadang-cadang spreads may vary from site to site, and disease incidence within the boundary of spread is variable (Zelazny and Niven, 1980; Zelazny and Pacumbaba, 1982; Zelazny *et al*., 1982). Zelazny and Pacumbaba (1982) identified three beetle species that were more abundant in areas of high disease incidence, but no vector has been identified unequivocally.

Hanold and Randles (1997) analysed 1024 coconut samples from an extensive survey for the presence of viroid-like sequences that shared sequence similarity with CCCVd. Only India had no samples which tested positive for such sequences. In some countries in Southeast Asia, a high incidence of CCCVd-like sequences has been detected. Further characterization of the newly detected molecules should determine whether any of them warrant classification as new viroids, and the health risks, if any, that these molecules pose to coconut and other palms.

Management

Replanting reduces disease losses since the rate of spread in new plantings is not related to the closeness of affected palms (Bigornia, 1977). Since removal of symptomatic palms does not eradicate cadang-cadang, Randles *et al*. (1997) suggested that removing all

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palms that were positively indexed for CCCVd infection might be more successful.

Trials have been established with nuts from palms that have remained healthy in areas with a high incidence of cadang-cadang in the hope of finding tolerance. Palms from these trials are being tested by both natural and experimental inoculation. Also, in directed breeding, dwarf × tall crosses are being made to produce hybrid seed that are challenged by experimental inoculation with the viroid.

Mild strain cross-protection is used for a number of economically important viruses, and the lack of options to control cadang-cadang has raised hope that this strategy could work for this disease. Rodriguez and Randles (1993) found variations in the pathogenicity and conserved domain of CCCVd associated with the 'brooming' form of the disease, suggesting that other variants may still be found. Similarly, Rodriguez (1993), using molecular detection techniques, found variants of CCCVd in ginger (*Alpinia* sp.) and arrowroot from the Philippines (Fig. 8.4).

Coconut foliar decay

Coconut foliar decay, which is also known as New Hebrides coconut disease, affects introduced coconut palm varieties on, and is limited to, the island of Vanuatu.

Symptoms

The local 'Vanuatu Tall' variety remains symptomless after infection, but introduced varieties and progeny derived from hybridization between these varieties and 'Vanuatu Tall' develop a range of symptoms (Calvez *et al*., 1980). These begin with partial yellowing of one or more of the 7th to 11th oldest leaves that spreads the length of the leaf and to adjacent leaves in the whorl. A lateral necrosis of the petiole then develops, which causes the leaf to die and hang from the canopy. This appearance of a normal apex and several yellowish leaves followed by death of young leaves that hang through green older leaves is characteristic of this disease (Randles *et al*., 1999). Narrowing of the trunk can occur but, with symptom remission, as occurs in tolerant varieties, it may thicken once again. Susceptible varieties die between 1 and 2 years after symptoms first appear.

Causal agent

Foliar decay is caused by a small, 20 nm in diameter, isometric virus particle, *Coconut Foliar decay virus* (CFDV) (Randles and Hanold, 1989). It is classified as a nanovirus and contains a 1.76 kb genome of circular single-stranded DNA (Randles *et al*., 1987). Purified preparations of CFDV or sap from infected coconut palms are not infective.

Epidemiology

CFDV is transmitted in nature by the planthopper, *Myndus taffini* (Julia, 1982). *M. taffini* breeds on the roots of a native tree species, *Hibiscus tiliaceous*, which grows prolifically on Vanuatu. It is a common reservoir for the vector in and around coconut plantations, but it has not been shown to host the virus. No other plant hosts of CFDV have been reported.

The development of molecular techniques for detection, such as polymerase chain reaction (PCR) (Rohde *et al*., 1990; Randles *et al*., 1992) and a cDNA probe (Randles *et al*., 1999), have allowed the diagnosis of CFDV in palms in the absence of symptoms. This led to the discovery that some 'Vanuatu Tall' palms have symptomless infections. Typically, CFDV can be detected in young leaves and secondary roots where it is localized in vascular tissues, primarily phloem, in a non-systemic pattern (Randles *et al*., 1992). CFDV can be detected in coconut embryos and husks, but not pollen (Randles, 1997). Although seed transmission of foliar decay has not been recorded, movement of nuts from infected palms could result in the transfer of CFDV to new sites.

Management

The local 'Vanuatu Tall' and 'Vanuatu Red Dwarf' coconuts are tolerant and offer the best means of controlling the disease (Persley, 1992).

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Ganoderma butt rot

Ganoderma butt rot, which is also known as basal stem rot, is a lethal disease of coconut and other species of palm. In Florida (USA), Elliott and Broschat (2001) listed 57 palm taxa that were affected. Although the disease has its greatest economic impact on oil palm, *Elaeis guineensis*, it can also damage coconut palm in the landscape and in plantations. For example, in India, losses of up to 31% have been reported (Bhaskaran and Ramanathan, 1984), and complete plantations were destroyed within 7–8 years when control measures were not used (Bhaskaran, 2000).

Symptoms

Overall symptoms vary from region to region, but the one found in all areas is internal discoloration of lower stem tissue. Initially, older fronds begin to yellow, droop and appear drought stressed (Peries, 1974). They eventually die, collapse parallel to the trunk, and either remain attached to the tree or break off. In some regions, younger and younger leaves become pale green and chlorotic, and the size of new leaves decreases. Several unopened leaves may develop in the crown, and flowering and nut set decrease and then stop. Eventually, only a few stunted leaves remain in the canopy before the palm dies. From the onset of symptoms to death usually takes 1–4 years.

In India, stem bleeding (exudation of a reddish brown liquid), chocolate brown bands that extend up the trunk and extensive root rot may also occur (Bhaskaran, 2000). These symptoms are not found in all regions and may reflect differences among the different causal agents. Stem bleeding may extend to 10–15 m up the trunk, but the pathogen usually extends no higher than 1.5 m (Elliott and Broschat, 2001). Internally, colonized tissues are discoloured brown.

Basidiocarps of the pathogens may form on lower portions of the trunk at any stage in disease development, and may appear before external symptoms develop. Basidiocarps also develop on stumps of trees that were killed by the disease or by other causes.

Causal agents

The taxonomy of the causal agents is confused (Seo and Kirk, 2000). Traits that have been used to delineate species of *Ganoderma* often are ambiguous and can vary when a given species is observed in different environments. New species have been named based on one or a few specimens, and hundreds of these have been reduced to synonymy with well-established species upon further examination. Furthermore, assignment of isolates to a given species has often been indiscriminate, resulting in erroneous host ranges being published for a given species. Although different taxa appear to affect coconut palm, with one exception their identities are unclear.

G. zonatum is reported from Florida, South America, Java and tropical Africa (Steyaert, 1975b), and the name has also been used for collections from Southeast Asia and Oceania (P. Bridge, Royal Botanic Gardens, Kew, personal communication, 2002). It affects mainly palms, but has been reported on *Eucalyptus* spp. and as a rotter of dead wood. The fungus produces sessile, reniform to irregular, dimidiate basidiocarps up to 30–40 cm at their widest point and up to 9 cm thick at the base (Fig. 8.7) (Steyaert, 1975b; Ryvarden, 2000). The upper surface usually is tuberculate and sometimes sulcate, glossy (laccate) and reddish brown, but darkens with age. The lower pore surface is white. In cross-section, the cutis is 20 μm thick and underlined in yellow. The context extends through one-third to half of the depth of the bracket and is tan, whereas the lower tube layer is walnut brown and half to two-thirds the bracket thickness. Basidiospores are pale yellow, ellipsoid, with long and thick echinules, and 9–16 × 5.5–9 μm. Basidiocarps from previous years that remain on palms are infertile.

G. boninense occurs in Australia, Japan, Indonesia, Malaysia, the Philippines, Samoa, Sri Lanka and Tasmania (Steyaert, 1975a). Its host range includes mainly palms, but some dicot tree species as well. It causes basal stem rot of oil palm in Southeast Asia and Oceania (Abdullah, 2000). Although it is isolated commonly from coconut palms, it does not cause disease. Coconut and *G. boninense* appear to have co-evolved in the region to

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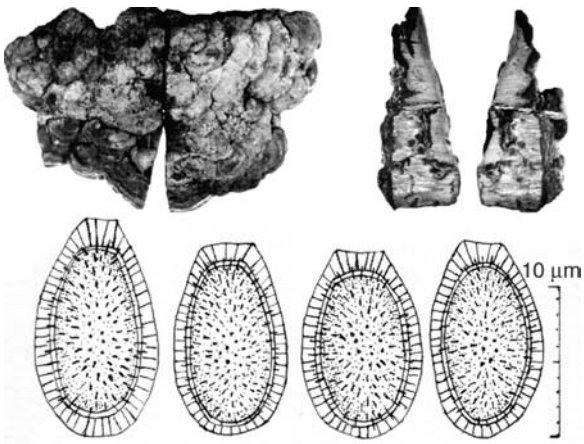


Fig. 8.7. Clockwise from top left-hand corner basidiocarp, sections of a basidiocarp and basidiospores of *Ganoderma zonatum* (from CMI description no. 448).

reach a state of benign equilibrium (J. Flood, CABI, personal communication, 2002).

A *Ganoderma* sp. kills palms in Sri Lanka and India (Peries, 1974; Bhaskaran, 2000). It has been called variously *G. applanatum*, *G. boninense* and *G. lucidum*, but resembles none of these species. Although it has a narrow genetic base, its true identity and affinity for these or other species are not known (P. Bridge, Royal Botanic Gardens, Kew, personal communication, 2002). Work to clarify its place in the genus is needed.

Other species of *Ganoderma* that have been reported on palms in general or specifically on coconut palm include *G. cupreum*, *G. miniatotinctum*, *G. tornatum* and *G. xylonoides* (Moncalvo, 2000). Their validity as species and importance and identification on coconut palm are uncertain.

Epidemiology

G. zonatum is thought to move primarily via basidiospores, but there are no experimental data to support this assumption (Elliott and Broschat, 2001). Recent work on another laccate species, *G. boninense*, indicates that basidiospores probably play a major role in disseminating it, either by directly infecting the host or by indirect infections from organic debris or dead host materials that they have colonized (Miller *et al*., 1999). Somatic incompatibility tests demonstrated the presence of highly diverse populations in plantations to the extent that it was uncommon to find two isolates of the same somatic incompatibility group. Based on these results, root-to-root contact between palms, an avenue that previously was thought to be important, was concluded to be of no consequence. It is assumed that basidiospores infect wounds on trunks or cut ends of palm fronds, but experimental evidence is lacking.

Management

In general, this is a disease of palms with a woody trunk. Mortality begins at ~10 years of age (Venkatarayan, 1936). Where the disease is established, sanitation is most important. Basidiocarps and infected trees should be removed and destroyed whenever they are found. All root systems, stumps and trunks of palms that die from other causes should also be removed as quickly as possible to prevent their colonization by these pathogens. Wounding should be avoided.

Fumigation of infested sites may be beneficial (Simone, 1994a). Products containing dazomet, metam sodium or methyl bromide+chloropicrin are effective. Although treating stumps with creosote, copper-based fungicides and tridemorph did not inhibit colonization of stumps, urea appeared to be effective and enhanced microbial decay of stumps in Sumatra.

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Grey leaf blight

Grey leaf blight was first recognized in Guyana in 1931 and is now known throughout the tropics (Cook, 1971; Holliday, 1980). It usually is a minor problem, but can be severe under crowded or wet conditions, or after leaves are damaged by insects. Young palms in nurseries are especially vulnerable.

Symptoms

Symptoms begin as small yellow to brown spots on leaflets and raches that develop grey centres with dark brown borders as they enlarge. Dark, globose acervuli of the causal fungus form in lesion centres on upper leaf surfaces (Plate 60). Lesions elongate parallel to veins, and may eventually coalesce to form large, irregular necrotic areas on leaves. In severe cases, leaflet tips and margins become grey and frizzled, imparting a blighted appearance to the canopy.

Causal agent

Pestalotiopsis palmarum causes grey leaf blight (Mordue and Holliday, 1971). Although a teleomorph in the genus *Rhynchosphaeria* has been reported for the pathogen in India (Agnihothrudu and Barua, 1965), it apparently is uncommon.

Conidiophores are hyaline, cylindrical to obovoid, 1–4 μm in diameter and 5–18 μm long (Fig. 8.8). Conidia emerge from acervuli in black cirri, are fusiform, usually straight, four-septate with slight constrictions at septa, and 20 (17–25) × 6 (4.5–7.5) μm. Three median cells in conidia are olivaceous and larger than the hyaline apical and basal cells. Apical cells have three (rarely two or four) cylindrical appendages that are 16 (5–25) μm in length. Basal appendages are straight and 2–6 μm long.

On potato dextrose agar (PDA), colonies initially are white and fluffy. Acervuli develop in yellowish clumps and produce greenish black masses of conidia. Colonies are usually diurnally zonate and, on the obverse side, exhibit little coloration.

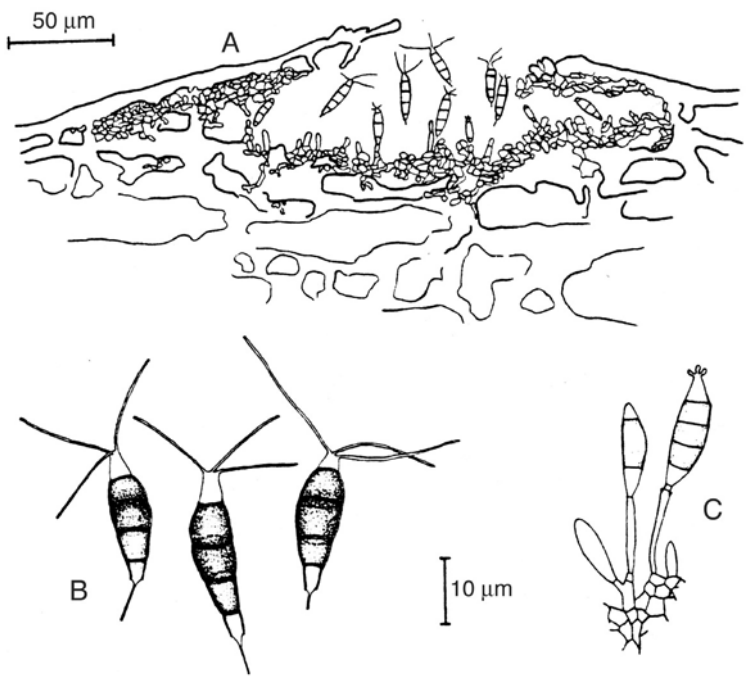


Fig. 8.8. (A) Acervulus, (B) conidia and (C) conidiophores of *Pestalotiopsis palmarum* (from CMI description no. 319).

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Epidemiology

The pathogen does not exhibit much host specificity, and several species of palm, as well as some non-palm hosts, are affected. *P. palmarum* is a relatively non-aggressive pathogen, and young, wounded or otherwise weakened host tissues are infected preferentially, presumably by conidia. The disease develops slowly on palms, and symptoms are usually only seen on older leaves. Thus, older leaves and infested host debris are the primary sources of inoculum.

Infection events apparently have not been studied in coconut palm. However, in banana, the fungus was shown to form appressoria and infection pegs that penetrated host surfaces directly (Vakili, 1963). Rainfall and high humidity play important roles in the development of this disease.

Management

Control measures are usually needed only in nurseries. To prevent damage from occurring in these situations, optimal growth of young plants is required. In particular, insect damage should be suppressed, and overcrowded or unsanitary conditions should be avoided. If possible, dry foliage should be maintained.

In nurseries, or the rare instances where control in mature trees is required, broad-spectrum, protectant-type fungicides are efficacious.

Hartrot

Hartrot is a fatal wilt disease. Historically, it has also been known as bronze leaf wilt, fatal wilt, Coronie wilt, Cedros wilt or Marchitez. Hartrot was first recognized during the early 1900s in Surinam where it now prevents successful coconut production. During the mid-1970s, the disease killed an estimated 15,000 coconut palms in the Cedros region of Trinidad within 3 years. Currently, it appears to be restricted to Central America (Costa Rica, Honduras and Nicaragua), South America (Brazil, Colombia, Ecuador, Guyana, Peru, Surinam and Venezuela) and the West Indies (Grenada, Trinidad and Tobago).

Symptoms

The earliest symptom on coconut is a simultaneous yellowing or browning of the lowest two or three leaves, starting from the tips and spreading to the leaf base (Plate 61). Immature nuts are shed but mature nuts generally remain attached for longer. As discoloration advances to involve younger leaves, older leaves become necrotic, desiccate and may break near the petiole. The rachillae of newly opened and unopened inflorescences are partially rotten and blackened, while leaves of the upper crown are still green; they die and collapse when the disease reaches an advanced stage. Roots also show apical or complete necrosis and desiccate at this time. Finally, when most of the leaves are brown and nut drop is complete, a putrid wet rot of the basal spear leaf and apical meristem develops and the palm dies.

Coconut palms of all ages are equally susceptible, as are both tall and dwarf types. In general, symptoms are indistinguishable from those of red ring and lethal yellowing. Mortality occurs within 1–3 months of the onset of external symptoms, depending on the age, size and condition of the palm, and vigorously growing palms succumb most rapidly.

Causal agent

Hartrot disease is associated with the systemic colonization of coconut palm by a culturable, phloem-restricted, uniflagellate protozoan (phylum *Euglenozoa*, order *Kinetoplastida*, family *Trypanosomatidae*) (Fig. 8.9). Plant-associated trypanosomatids belong to the genus *Phytomonas*, which includes flagellates that parasitize latex cells of laticiferous plants, and intraphloemic flagellates that infect non-laticiferous host plants. Trypanosomatids from laticiferous plants are not considered to be pathogenic, although some damage fruit (Dollet, 1984). Speciation of those from non-laticiferous host plants has not been fully resolved, nor has the pathogenicity been demonstrated for those that are associated with coconut. Their role as phytopathogens is supported

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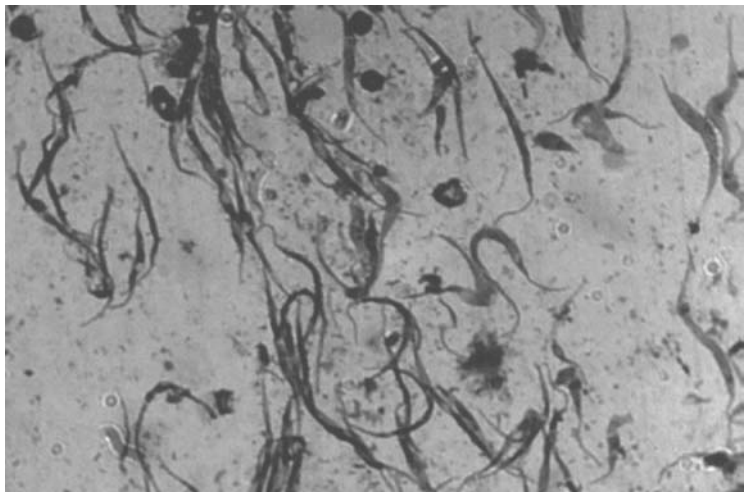


Fig. 8.9. Uniflagellate protozoans (phylum Euglenozoa, order Kinetoplastida, family Trypanosomatidae, genus *Phytomonas*) in the sap of a coconut palm affected by hartrot (photo: Michel Dollet).

by their consistent association with hartrot symptoms, and the correlation between the level of phloem colonization by *Phytomonas* and disease severity (Waters, 1978).

Although culturable, *Phytomonas* isolated from palms initially require several passages on complex media containing cultured insect cells before they can be grown in cell-free media (Menara *et al*., 1988). The promastigote form predominates in coconut. These flagellates are between 1 and 1.5 μm wide and 12 and 27 μm long, including a single, 7 μm long flagellum. Within the cytoplasm, the base of the flagellum is associated with a kinetoplast composed of a reticulate network of electron-dense DNA. The ribosome-rich cytoplasm also contains a single nucleus and both electron-translucent and -dense, membrane-bound structures (Waters, 1978).

Isolates from palms are morphologically uniform. However, they can be distinguished by: their isoenzyme profiles (Guerrini *et al*., 1992); the sizes, homology and restriction length polymorphisms of minicircle components of their kinetoplastid DNA (Muller *et al*., 1995); the presence of double-stranded RNA or virus-like particles (Marche *et al*., 1993); and the sequences of their small subunit 18S rDNA (Marche *et al*., 1995).

Epidemiology

The epidemiology of hartrot is poorly understood. Although it is capable of rapid spread, as evidenced by the epiphytotic in the Cedros region of Trindad (Waters, 1978), spread is usually slower and more localized. Hartrot affects mostly bearing palms, although very young palms, 18–20 months old, are sometimes involved. To date, all coconut ecotypes that have been tested have been equally susceptible (Louise *et al*., 1986). When symptoms first appear, 10–100% of mature sieve tubes may contain flagellates and many are completely occluded (Parthasarathy *et al*., 1976), presumably obstructing transport of photosynthates. Sap expressed from coconut inflorescences or roots at this stage contain flagellates that can be resolved by light microscopy (Fig. 8.9) (Waters, 1978).

Several species of the pentatomid insect genera *Lincus* and *Ochlerus* are known to transmit the trypanosomatids responsible for hartrot (Desmier de Chenon, 1984; Dollet, 1984). Crepuscular feeders that do not fly, adults apparently acquire *Phytomonas* during feeding activity on diseased palms and vector it to adjacent palms. Other palms, such as *Bentinckia nicobarica*, *Maximiliana maripa* and *Roystonea regia*, may serve as alternative sources of the protozoan (Dollet, 1984; Kastelein and Parsadi, 1986).

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Management

Control of hartrot relies primarily upon early detection, prompt removal and destruction of affected palms. Treatment of palms with insecticides such as lindane or deltamethrin arrests disease spread (Louise *et al.*, 1986), but whether this strategy is useful or practical over the long term is unproved. Technical guidelines have been established to prevent the inadvertent spread of hartrot during the transfer of coconut germplasm (Frison *et al.*, 1993).

Leaf scorch decline

Leaf scorch decline was first reported in southern Sri Lanka in 1955 (Kirthisinghe, 1966). However, there is evidence that it existed in the area considerably before this time (Bryce, 1921; Park, 1927). The disease continues to spread and is now found in almost all coconut-growing areas of the country (Fernando, 1999).

Symptoms

Scorching and drying of leaflets accompanied by a slight curling of the lowermost leaves are characteristic of the disease. Scorching gradually extends from the tips of leaflets to the leaf midrib and then progresses to successively younger leaves in the upper crown. On each leaf, distal leaflets are more severely affected than proximal ones. The oldest affected leaves usually remain attached to palms for an extended time as compared with leaves of similar age on healthy palms (Ekanayake, 1963). Tapering of the trunk just below the crown is apparent on palms that have reached mid-stage development of the syndrome. As symptoms advance, fewer and smaller inflorescences are produced and nuts are elongated in shape. In its advanced stage, new leaves are markedly reduced in length and width, and palms become completely unproductive as the crown diminishes in size and eventually topples from the trunk (Mahindapala and Chandrasena, 1975; Rajapakse and Fernando, 1997).

Foliar symptom development is accompanied by considerable root decay. Root systems of affected palms are more fibrous than those of healthy palms due to rootlet necrosis. Cross-sections of root tissues reveal blackening of cortical tissue and blockage of xylem vessels by tyloses (Peries, 1968; Jayasekara *et al.*, 1990). No consistent changes in trunk tissues have been reported. The full course of the syndrome averages between 5 and 10 years. Similar declines have been observed on several other palm species (Davis, 1962).

Epidemiology

An initial 1963 survey of the Southern Province of Sri Lanka revealed a disease incidence of 5.1%. Subsequent surveys of the major coconut-growing areas during 1985 and again in 1996 showed an increase of up to 12%, and incidence was highest in small holdings (<1 ha). The distribution pattern of affected palms was scattered, and there was no indication of focal origin or directional spread (Fernando, 1999). At this time, no cause has been identified for this disease.

Lethal bole rot

Lethal bole rot, caused by the basidiomycete *Marasmiellus cocophilus*, is an important disease of coconut palm in East Africa but is rarely found outside this region. The disease is often fatal to palms up to 8 years old. It spreads readily through the soil by root contact between palms and probably by airborne basidiospores. The first visible symptoms are a wilting of the oldest leaves that turn yellow or bronze. At an advanced stage, reddish dry rot and cavities lined with fungal mycelium appear in the tissue of the basal trunk (bole). Sometimes the entire bole region of palms is decayed due to invasion by secondary bacteria. Extensive destruction of the root system, death of the spear leaf and a progressive soft rot of the apical bud accompany basal trunk decay.

Seed infestation by *M. cocophilus* leading to a basal stem rot of seedling coconuts has also been described (Jackson and Firman,

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1982; Jackson and McKenzie, 1988). Affected seedlings in field nurseries show premature death of the oldest two or three leaves, and often snap at the junction of the stem and nut. Younger leaves develop a brown rot at the leaf bases. Sporocarps of *M. cocophilus* are produced in abundance from seed husks and bases of dead leaves. After transplanting, leaves of surviving seedlings unfurl before becoming fully emerged. Decay of some areas of the bole and roots is usually evident. In some instances, affected seedlings eventually recover to produce normal roots and leaves.

M. cocophilus has been reported from several grasses including goosegrass, marsh grass and Bermuda grass (Frison *et al.*, 1993). They may serve as alternative hosts and sources of inoculum.

Lethal yellowing

The term lethal yellowing (LY) was first used in the mid-1950s to denote a fatal disease of uncertain aetiology that had affected coconut palms in western Jamaica since the 1800s (Nutman and Roberts, 1955). LY is found in the Bahamas, the Cayman Islands, Cuba, Dominican Republic, Haiti, Jamaica and the southern USA (Florida and Texas), and recently has spread to southern Mexico, Belize and Honduras (Ashburner *et al.*, 1996; Eden-Green, 1997a). It threatens global coconut production because it spreads rapidly, kills palms quickly and is incurable. About two-thirds of the world's coconut palms are susceptible (Harries, 1978).

Symptoms

Symptoms on highly susceptible mature, tall-type coconut palms begin with a premature drop of most or all the fruit. Aborted fruit often develop a brown–black, water-soaked appearance at the calyx end. Inflorescence necrosis follows, with flower stalks partially or totally blackened as they emerge from the surrounding spathe, in contrast to the creamy white to yellow flower stalks on healthy palms (Plate 62). The foliar yellowing phase initially involves the lowest, older leaves and eventually encompasses the entire crown (Plate 63). Occasionally, this symptom is first seen as a single yellow leaf (flag leaf) in the mid-crown. Yellowed leaves remain turgid before turning brown, then desiccate and hang down around the trunk for several days before falling. Death of the apical bud occurs midway or late in the foliar yellowing phase, at which time root necrosis is extensive. Bud death is accompanied by a putrid decay of these tissues, due to the invasion of secondary microbes, and collapse of the newly emerged spear (youngest) leaf that then hangs from the crown. Finally, the remaining crown withers and topples from the palm, leaving a bare trunk standing. Palms succumb to LY within 3–6 months of symptom onset.

On dwarf-type coconuts, symptoms also include premature fall of nuts and inflorescences followed by a pronounced browning of older leaves, rather than the yellowing that is typical on most tall varieties. Often, foliar symptoms on the 'Malayan Green Dwarf' more closely resemble those of a wilt during which inflorescences shrivel inside unopened spathes and fail to emerge. Pinnae of mid- and upper-crown leaves are noticeably flaccid and fold inward, and pronounced constrictions in the trunk may develop during the later stages of the disease.

Causal agent

Phytoplasmas (previously known as mycoplasma-like organisms or MLOs) were implicated as the probable cause of LY when these minute, wall-less prokaryotes were first observed by electron microscopy in phloem sieve tubes of symptomatic, but not healthy, coconut palms in the early 1970s (Plavsic-Banjac *et al.*, 1972). Unlike other phytopathogenic mollicutes, phytoplasmas are obligate parasites. Thus, evidence for their being the cause of LY is derived primarily from their consistent association with diseased palms, from symptom remission in palms after injection with tetracycline but not penicillin antibiotics, and from the absence of other pathogens.

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Besides coconut, the LY phytoplasma affects at least 35 other palm species and the arborescent monocot *Pandanus utilis* (Harrison and Oropeza, 1997). In palms, phytoplasma profiles vary from ovoid to filamentous when viewed by electron microscopy. In coconut, non-filamentous forms average 295 nm in diameter, and filamentous forms average 142 nm in diameter and at least 16 μm in length (Waters and Hunt, 1980). Phytoplasma cells are enclosed by a trilaminar unit membrane and contain cytoplasm with DNA fibrils and ribosomes (Thomas, 1979; Thomas and Norris, 1980).

Molecular diagnostic assays that employ DNA probe hybridization and the PCR have been developed for the LY phytoplasma (Harrison *et al*., 1992, 1994). Besides enhancing pathogen detection and identification in host plants or vector planthoppers, these assays have also shown that the phytoplasma exists as a group of closely related, possibly identical strains in the western Caribbean (Harrison and Oropeza, 1997). They are phylogenetically distinct from other coconut-infecting phytoplasmas in Africa (Tymon *et al*., 1998), and are most closely related to a phytoplasma associated with decline-type diseases of coconut and the arborescent monocot, *Carludovica palmata*, in southern Mexico, and Canary Island date palm, *Phoenix canariensis*, in Texas (Harrison *et al*., 2002).

Epidemiology

The onset of external symptoms follows a protracted incubation (pre-symptomatic) phase estimated as between 114 and 405 days (McCoy *et al*., 1983). Generally, phytoplasma concentrations are highest in immature rather than mature tissues and may be found most reliably in phloem-rich unemerged leaf bases surrounding the apical meristem of affected hosts. They are less abundant in flower stalk bases or roots and are only occasionally present and sporadically distributed in mature leaves.

From primary disease foci that typically involve a few palms, two types of disease spread have been described. One involves spread in a random pattern around the focus, eventually claiming most of the susceptible palms within, and a second type occurs as a series of jumps of a few to 100 km or more to establish new disease foci from which the local pattern of spread is then repeated. McCoy *et al.* (1983) studied disease spread and apparent infection rates during the epidemic of LY in southeastern Florida during the early 1970s. After an initial lag phase, the number of diseased palms increased in a logarithmic fashion. It was estimated that each affected coconut palm in a primary focus was responsible for the infection of 4.6 new palms during the first 8 months of the epidemic, and 9.3 new palms within 2 years.

In Florida, the rates of spread generally have been lower in coastal than in inland sites. Although a higher level of cultural maintenance occurs at the inland sites, the actual reason for this relationship is unclear. Differences in the rate of spread at different geographical locations have also been noted. In 3 years, the disease spread over 128 km in southeastern Florida, and similar rates of dispersal have been reported in southern Mexico (Oropeza and Zizumbo, 1997). In contrast, LY required ~60 years to move across Jamaica, a distance of 238 km.

There is persuasive evidence implicating a planthopper, *Myndus crudus*, as a primary vector of LY (Fig. 8.10). In Florida, transmission of LY to coconut and other susceptible palm species occurred following their exposure to *M. crudus* collected from palms in areas with a high incidence of LY, but did not occur when palms were protected from the planthopper (Howard *et al*., 1983, 1984). Furthermore, *M. crudus* and LY have coincident distributions in the Americas (Howard, 1983).

Management

Given our present understanding of the disease, use of resistant ecotypes and hybrids offers the only practical long-term solution to LY (Harries, 2001). Although a recent report (Broschat *et al*., 2002) questions whether the 'Malayan Dwarf' and hybrid 'Maypan' ('Malayan Dwarf' × 'Panama Tall') possess high levels of resistance, they

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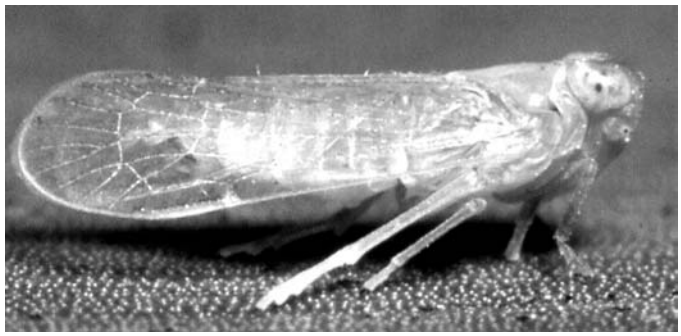


Fig. 8.10. *Myndus crudus*, the probable vector of lethal yellowing in the Americas (photo: J. De Filippis).

have been used extensively for replanting in Jamaica and other countries that are affected by the disease. Promising levels of resistance have been identified in other coconut ecotypes that include 'Chowghat Green Dwarf', 'Fiji Dwarf', 'Red Spicata Dwarf', 'Sri Lanka Yellow Dwarf' and 'King' (Harries, 1995), but these ecotypes have not been exploited commercially.

To discourage the inadvertent spread of LY, commercial movement of living palms and palm seeds from LY-affected to disease-free areas is generally not permitted. However, quarantine requirements vary according to the localities that are involved. Technical guidelines for the safe movement of coconut germplasm from LY-affected areas for research but not for commercial purposes have been developed under the auspices of the International Board for Plant Genetic Resources (now known as the International Plant Genetic Resources Institute) (Frison *et al*., 1993). Other measures, such as the eradication of affected palms, have not reduced the spread of LY, but slight reductions were achieved by insecticide suppression of the reputed vector (Howard and McCoy, 1980). Prophylactic antibiotic treatments, although effective (McCoy *et al*., 1976), are not economical in commercial coconut production.

Lethal yellowing-like diseases

Recurrent LY-like diseases also affect coconut cultivation in the eastern hemisphere. These include Awka disease (Nigeria); Cape St Paul wilt (Ghana) (Plate 64); Kaïncopé disease (Togo); Kribi disease (Cameroon); lethal decline (Tanzania, Kenya and Mozambique) (Plate 65); Kalimantan wilt, Natuna wilt (Fig. 8.11) and Sulawesi yellows in Indonesia; leaf scorch decline in Sri Lanka; Malaysian wilt in peninsular Malaysia; and Tatipaka disease and root wilt in India (Eden-Green, 1997b). The symptoms and epidemiology of, and devastation that is caused by, these diseases closely resemble those of LY (Plates 64 and 65; Fig. 8.11) (Eden-Green, 1997b). However, it was not until molecular diagnostics became available that the associated phytoplasmas could be easily diagnosed (Tymon *et al*., 1997) and relationships between the African and Caribbean LY diseases established (Tymon *et al*., 1998; Mpunami *et al*., 1999).

Awka wilt, Cape St Paul wilt, Kaïncopé and Kribi

These diseases spread rapidly and begin with premature nutfall. Varietal susceptibility has been studied most extensively in Ghana, where natural exposure of germplasm in replicated field trials has demonstrated that the 'Vanuatu Tall' and 'Sri Lankan Green Dwarf' varieties are most tolerant. Insect vectors of these diseases have not been identified. However, in Nigeria, two insects, *Meenoplus proximus* and *Malenia cocos*, were widespread on palms year round and considered the most promising candidate vectors of Awka wilt (Aisagbonhi and Kolade, 1994). *Myndus adiopodoumeensis* has been investigated intensively as a potential vector of St Paul wilt in Ghana, but no posi-

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Fig. 8.11. Decline of a coconut plantation in Indonesia caused by Natuna wilt. Note the presence of healthy, non-bearing palms in the foreground and the absence of nuts on palms in the background (photo: P. Jones).

tive transmissions using this planthopper have been demonstrated to date (Eden-Green, 1997b).

Molecular studies have shown that phytoplasmas implicated as aetiological agents of coconut diseases in Ghana and Nigeria are distinct from, although related to, the LY phytoplasma (Tymon *et al*., 1998).

Coconut lethal disease

Lethal disease (LD) was first identified in Tanzania, but subsequently has also been found affecting coconuts in the adjacent countries of Kenya and Mozambique. In Tanzania, disease incidence differs significantly from region to region. In the southern region, LD has killed >50% of the coconut palms since 1965, while in the northern region it has killed only 8.5% (Schuiling *et al*., 1992). Using molecular diagnostics, Mpunami *et al*. (1999) showed the presence of two distinct phytoplasma strains amongst diseased East African coconuts. The most prevalent of these has been classified as the LD phytoplasma (Tymon *et al*., 1998), while the second strain is most closely related to the phytoplasma associated with Cape St Paul wilt in West Africa.

Studies in Tanzania over many years have shown that coconut varieties resistant to LY in the Caribbean are highly susceptible to LD. Both Schuiling *et al*. (1992) and Kullaya *et al*. (1997) concluded that no exotic varieties have exploitable resistance to LD. However, the 'Pemba Red Dwarf' as well as a local tall ecotype show promising levels of resistance. Most recently, *Diastrombus mkurangai* and *Meenoplus* spp. (Homoptera: Meenoplidae) have been implicated as vectors of LD in Tanzania (Mpunami *et al*., 2000).

Kalimantan wilt

This disease presently is restricted to the district of Kotawaringin Timur of Central Kalimantan where it threatens at least 17,000 ha of coconut grown by smallholders in settlements forming a 5–7 km wide strip along the Sampit river. Symptoms in the area of Samuda resemble those of Natuna wilt. Immature and middle-aged nuts already present on palms may be malformed and are often elongated in shape with poorly developed endosperm (Plate 66). Death of the palm occurs 6–18 months after symptoms first appear. The disease first occurs as a focus within a planting and then spreads

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outward to neighbouring palms. Some palms may escape infection as new foci develop some distance away. No systematic epidemiological study of Kalimantan wilt has been undertaken, but these preliminary observations suggest the involvement of an airborne vector (Warokka, 1999).

Jones *et al.* (1999) implicated the involvement of phytoplasmas with palms affected by Kalimantan wilt and Natuna wilt based on evidence obtained from diagnostic PCR assays. No comparative studies have been carried out as yet to determine the relationship between these putative aetiological agents and other phytoplasmas from coconut palm.

Natuna wilt

Lying 4° north of the equator between Peninsular Malaysia and Borneo, Natuna Besar is the largest of the Natuna Islands with an area of 1095 km^2. Natuna wilt was first recorded in 1965 in Kelarek in the Western District (Sitepu, 1983). This lethal disease has now spread throughout Natuna Besar and the offshore islands to the southwest (Warokka, 1999). Although Natuna wilt usually affects palms that are >10 years old, symptoms have been observed on palms as young as 7 years. Non-bearing palms apparently are unaffected (Fig. 8.11).

Symptoms first appear as a pronounced wilting of the lower leaves that progresses to involve successively younger leaves. Affected leaves droop and lose their sheen before changing from green to brown, without any pronounced yellowing stage as seen with LY. While necrosis of immature inflorescences does not occur, a cessation of nut production coincides with the onset of foliar symptoms (Fig. 8.11). Immature and middle-aged nuts already present on palms may be malformed and often are elongated, with poorly developed endosperm. Other nuts may fail to develop any endosperm and consist of solid fibrous husk only. Nutfall is inconsistent and therefore not a reliable disease symptom. An internal examination of bud tissues reveals no apparent abnormality of these immature tissues during the early stages but, as symptoms progress, brown lesions appear on both the basal region of the unemerged petioles and inflorescence sheaths.

According to a 1980 disease survey, as many as 500,000 palms were affected on Natuna Besar (Sitepu, 1983). In 1997, it was estimated that the regionwide incidence of Natuna wilt was ~6–7%, although the incidence within certain localized sites approached 40% (Warokka, 1999).

Premature decline

This disorder was first recognized during 1995 in the Western and North Western Provinces of Sri Lanka, but it may have been present there much longer. Palms >20 years of age are affected. In contrast to leaf scorch decline, palms die more rapidly, usually within 1–3 years of symptom onset. Symptoms begin as a slight drooping and pale green or yellowish green colour of the lower leaves. Due to the drooping, a noticeable gap forms between the lower and upper leaves of the crown. By this stage, an acute tapering of the trunk below the crown is evident. On palms with advanced symptoms, the size of the crown is greatly diminished as a result of a marked reduction in the length of newly formed leaves. Nut production ceases and is followed by death of the palm. The cause of this disease is not known.

Red ring

Red ring was probably first described in Trinidad in the early 1900s. Known locally as root disease or fever, it was eventually named red-ring disease by Nowell (1919) who noted the internal red-ring symptom in diseased palms in Grenada and Trinidad. Red ring is one of the most important diseases of coconut palm in the Neotropics. It is found in Central America, Dominican Republic, the Lesser Antilles, southern Mexico (Yucatan Peninsula) and parts of South America (Brazil, Colombia, Ecuador, French Guyana, Guyana, Peru, Surinam and Venezuela). Annual losses of 10–15% occur in some areas (Griffiths, 1987; Giblin-Davis, 1993).

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Symptoms

Symptoms on tall-type coconut palms include premature nutfall of all but mature nuts, necrosis and withering of inflorescences followed by yellowing, and senescence (browning) of progressively younger leaves (Plate 67). Leaf discoloration begins at the tips of the pinnae, moving inward to the rachis and basal petiole. Senescent leaves often break at or near the petiole and hang downward. A transverse section of the lower stem invariably reveals a diagnostic orange to red–brown ring 2–6 cm in width within 2–6 cm of the stem periphery (Plate 67). In longitudinal section, internal stem discoloration is usually continuous throughout its length as two discrete, peripheral bands that fuse in the basal stem while forming discontinuous streaks or spots in the upper stem. The apical meristem is not affected, while the normally white cortical tissues of roots turn orange to light red and desiccate. An internal red–brown discoloration that extends for varying distances into leaf bases may also be evident. Full development of internal symptoms occurs before the appearance of any recognizable external symptoms (Griffiths, 1987; Griffiths and Koshy, 1990).

Symptoms vary widely according to the environmental conditions, age and palm type. For example, in dwarf coconut ecotypes and dwarf × tall hybrids, leaves become brown and desiccate rather than turn yellow. Also, internal discoloration is brown rather than the orange or red–brown colour evident in most tall ecotypes. Severe structural damage to the crown of diseased palms results from larval feeding by the weevil vector, often causing the crown to topple over under its own weight. Coconut palms 3–10 years old are affected most frequently and die within a matter of months. Infection of older palms leads to a more protracted decline marked by the appearance of progressively smaller leaves and a cessation of normal fruit production (Hoof and Seinhorst, 1962).

Causal organism

The disease is caused by the red-ring or coconut palm nematode, *Bursaphelenchus cocophilus*, a stylet-bearing member of the order *Aphelenchida*. This obligate parasite is vectored to coconut and other palm species mostly by adults of the American palm weevil, *Rhynchophorus palmarum*, and also by the sugarcane weevils *Dynamis borassi* and *Metamasius hemipterus* (Giblin-Davis, 1993).

Both sexes of mature *B. cocophilus* are <15.5 μm in diameter and 775–1369 μm in length. Stylet length is ~11–15 μm in adults (Gerber *et al.*, 1989). Survival of the nematode depends upon the dispersal stage juveniles (third stage) that may persist intercellularly in coconut tissues for as long as 3 months (Griffiths, 1987). Third-stage juveniles are 700–920 μm long and have a pointed tail with or without a mucron. Up to 11,000 nematodes can be recovered per gram of discoloured stem parenchyma tissues, leaf petioles and roots of coconut, or from *R. palmarum* (Blair, 1964; Gerber and Giblin-Davis, 1990). Survivorship of *B. cocophilus* is poor in soil and water (<7 days), but has been prolonged experimentally on the mesocarp of near mature coconut fruits or in excised leaf stalks (Giblin-Davis *et al.*, 1989).

Epidemiology

Palm and sugarcane weevils colonize affected palms. Their larvae become associated with the dispersal stage of *B. cocophilus*, and carry it through their metamorphosis. Adult weevils then vector the nematodes to other palms (Giblin-Davis, 1993). The incidence of *R. palmarum* females that are infested with nematodes upon emergence from diseased coconut palms can be high. In a study in Trinidad, all were found to contain nematodes and >47% harboured >1000 nematodes (Gerber and Giblin-Davis, 1990). Seasonally abundant, *R. palmarum* is a strong flier capable of rapid long-distance travel. Weevils are attracted to moisture in the leaf bases of palms and by semiochemicals emitted from palm wounds (Giblin-Davis *et al.*, 1996). Visiting males produce 7- to 9-carbon methyl alcohol aggregation pheromones that attract weevils of both sexes to the palm for mating and oviposition (Giblin-Davis *et al.*, 1996). Transmission of *B. cocophilus* seems to occur primarily during oviposition. The nematodes are highly motile and only 10–50

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are needed in wounds for disease establishment (Goberdhan, 1964). Although nematodes are excluded from vascular tissues, occluding tyloses that develop in xylem vessels close to invaded tissues impair water uptake and translocation in palms (Giblin-Davis, 1993). In 4- to 5-year-old coconut palms, experimental inoculations with dispersal stage nematodes induced external symptoms in some palms within 28 days, with peak populations of nematodes occurring in tissues at 42 days. Test palms succumbed within 10 weeks (Goberdhan, 1964; Griffiths, 1987).

Management

Promising resistance has been identified in some coconut ecotypes (Dean, 1979) but has not been exploited. Disease control relies primarily on prompt removal and destruction of diseased or *R. palmarum*-infested palms. A rigorous phytosanitation programme coupled with mass trapping of *R. palmarum* using baits and a synthetic aggregation pheromone, Rhyncholure (racemic 6-methyl-2-hepten-4-ol), reduces weevil numbers and dispersion patterns while lowering incidence in oil palm plantations (Oehlschlager *et al*., 1995). Whether a similar strategy would be effective in coconut palm plantations is untested.

Root (wilt) disease

Root (wilt) disease first appeared in the late 1800s (Koshy, 1999). According to a survey conducted during 1984–85, the disease was present in the eight southern districts of Kerala, and in small isolated pockets in some northern districts and Tamil Nadu. Disease incidence varies considerably, and is highest in Kottayam (75.6%) and lowest in Thiruvananthapuram (1.5%). Both prebearing and mature palms are affected and, although the disease is not fatal, it causes estimated losses of 968 million nuts annually (Solomon, 1994). Although the disease is thought to be caused by a phytoplasma, studies on it have yielded conflicting results.

Symptoms

The primary symptom is an abnormal bending of the leaflets that is termed ribbing or flaccidity. Yellowing and marginal necrosis of leaflets of the outer whorls also occur. In seedlings and juvenile palms, foliar yellowing is uncommon and symptoms begin with a rotting of the spear leaf followed by leaf flaccidity. In some cases, flaccidity may precede spindle rot. Root necrosis is extensive in affected palms, and the capacity of the palm to produce new roots is drastically reduced. Shedding of buttons and immature nuts, a reduction in the number and size of leaves, and drying of the spathes and a necrosis of spikelets from the tip down on unopened inflorescences may also occur. Leaves break at the tips, then yellow, desiccate and collapse before falling off. Abnormal structural changes occur in vascular tissues. Tyloses are induced in xylem vessels of roots on diseased palms. In its advanced stages, the palm crown is markedly smaller, fewer or no inflorescences are produced, and flowering on juvenile palms is delayed.

Causal agent

There have been numerous studies to identify a causal agent, and fungi, bacteria, viruses and nematodes have all been implicated (Solomon, 1997). Phytoplasmas have been observed in transmission electron micrographs of sieve elements of some affected palms (Solomon *et al*., 1987). Further support for a phytoplasma aetiology was obtained following transmission of the disease by *Stephanitis typical*, a lacewing in the *Tingidae*. (Interestingly, all known phytoplasma vectors belong to genera of leafhoppers, planthoppers or psyllids (*Auchenorrhyncha*).) Furthermore, remission of symptoms occurred in 53% of the affected palms that were treated with oxytetracycline. The disease was also graft transmitted from coconut to periwinkle with dodder laurel (Sasikala *et al*., 1988). A serodiagnostic test (Solomon *et al*., 1987) and a physiological test based on stomatal resistance (Rajagopal *et al*., 1988) have been standardized for detecting the disease, but attempts to verify the presence of phyto-

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plasma in affected palms by molecular diagnostics such as PCR have not been successful.

Epidemiology

Spread is faster, although erratic, among palms grown in sandy loams or alluvial soils. Higher disease incidence has been observed in low-lying waterlogged areas bordering rivers and canals.

Management

Based on nutritional studies conducted at the Central Plantation Crops Research Institute (CPCRI) in Kerala, India, it is recommended that each affected palm be treated with 500 g N, 300 g P_2O_5, 1000 g K_2O and MgO annually to maintain its productivity. Viable economic yields can be obtained from diseased palms by spraying fungicides against leaf rot, adding of organic matter, raising green manure crops beneath trees, irrigating during summer months and adopting inter- and mixed-cropping and mixed farming (Nair *et al*., 1991; Solomon, 1994). Management practices advocated by CPCRI include eradicating severely affected palms and those that are affected prior to flowering.

Stem bleeding disease

Stem bleeding disease is found worldwide, and old plantings are affected most. Symptoms on the trunk begin as a soft, yellow decay (Simone, 1994b). Affected areas are dark and blacken with age. A reddish brown or rust-coloured liquid may ooze from infections, and may extend several feet down the trunk discolouring it black as it dries. Pith tissues of affected roots decay. Infections may coalesce and cause heart rot and, as the disease progresses, lower leaf pinnae may die and defoliation and death may occur.

The causal fungus, *Chalara paradoxa*, is described in Chapter 1. Its teleomorph, *Ceratocystis paradoxa*, is uncommon on palms. The pathogen infects primarily the trunk and roots through wounds. Chlamydospores of the fungus are important for its survival, and it is moved via rainsplash and air currents, as well as in infested soil by birds or rodents.

Wounding of the host should be avoided, and affected palms should be removed and destroyed (Simone, 1994b). When infections are restricted in size, they can be removed and treated with benomyl and a wood preservative.

Sulawesi yellows

A leaf yellowing disease of coconuts was reported from central Sulawesi in 1989. It has an unknown aetiology and is most prevalent in the areas of Donggal (Kecamatan Ampibabo) and Poso (Kecamatan Tojo). In these districts, ~175,000 ha of coconuts are grown on smallholder farms (Simatupang, 1999).

Typically, symptoms are first evident as a yellowing at the tips of leaflets on older leaves that progresses to the lamina. Yellowing progresses to involve successively younger leaves, but not to the spear, which remains green. Foliar discoloration may be accompanied by leaf necrosis. As new leaves are produced, they do not expand fully and remain small, imparting a stunted appearance to the crown. Nut production declines progressively. Nuts are small in size or misshapen. The syndrome affects both tall and hybrid varieties equally, but appears to be non-lethal (Simatupang, 1999). Its cause is not known.

Tatipaka disease

The disease first appeared in the Tatipaka village of Razole Taluk in the East Godavari District of Andhra Pradesh, India, after the cyclone in 1949, and has since been identified in the Srikakulam and Nellore Districts.

Symptoms

Tatipaka affects palms between 20 and 60 years of age. The onset of symptoms is characterized by sudden, profuse growth. Leaf production increases slightly, as does yield.

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The crown assumes a rosette appearance as leaves lengthen and become dark green. As this phase recedes, successive leaves are reduced in both size and number and become light green with chlorotic spots before slowly yellowing. In some cases, leaflets adhere together, resulting in a fasciated appearance. Expanded leaves often exhibit a bow-like appearance due to bending of the rachis; which may also twist. Senescent leaves remain attached and are found hanging below the crown instead of falling to the ground. Spathes are very small and contain few rachillae, and inflorescence production is progressively reduced. Fruit bunches contain both normal and atrophied nuts, or may be barren. Nuts become rounded with a soft mesocarp, whereas atrophied nuts consist only of husk. At an advanced stage, palms display extensive root rot and a reduced capacity for root regeneration. Tapering of the trunk and bending into an 'S' shape is common. Palms shed yellowed leaves in slow succession and may eventually die (Rethinam *et al*., 1989).

Causal agent

Ultrastructural examination of new roots, beneath the meristem, petioles of developing leaves and rachilla of new inflorescences has revealed the presence of phytoplasmas in phloem sieve tubes of some affected palms. Transmission electron microscopy observations have been corroborated by light microscopy using Dienes's stain, or fluorescent microscopy employing aniline blue as fluorochrome. Temporary remission of symptoms on diseased palms after treatment with tetracycline hydrochloride further supports the concept of a phytoplasma aetiology for the disease.

Management

Resistance has been noted in the dwarf cultivar 'Gangabondam', which is a good parent for the production of hybrids. Attempts to control the disease by foliar and soil application of various chemicals, hormones and nutrients were not successful. The primary control strategy is to arrest the disease's spread by identifying and roguing diseased palms. Eradication of 87% of the affected palms has been completed, and monitoring is continuing (Rajamannar, 1994).

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